



ExoIntel-Prime

Technical Guide and Scientific Methodology

Browser-based exoplanet transit modelling with archival photometry,
background computation, and WebGL visualisation

Author: Biswajit Jana
Project: ExoLight Transit Lab / ExoIntel-Prime
Document type: Technical guide and methodology note
Release focus: Browser-based transit photometry laboratory
Version: Public scientific guide for the compute-engine + WebGL release
Year: 2026

This document describes the scientific scope, numerical methodology, software architecture, interface logic, and interpretive limits of **ExoIntel-Prime**. The platform is intended as a research-grade educational and demonstrative environment for interactive exoplanet transit analysis, rather than as a fully validated inference pipeline.

Prepared for scientific documentation, technical review, and repository presentation.

Contents

1	Scope and Purpose	1
2	What ExoIntel-Prime Does	1
2.1	Who the site is for	1
2.2	What the site is not	2
3	Interface Overview	2
3.1	Launch screen	2
3.2	Main layout	2
4	Target Archive and Data Philosophy	2
4.1	Target archive fields	3
4.2	Observed versus synthetic targets	3
5	Core Transit Physics	3
5.1	Transit depth	3
5.2	Orbital phase	4
5.3	Scaled orbital distance	4
5.4	Impact parameter	4
5.5	Transit duration	4
6	Limb Darkening	5
6.1	Why limb darkening matters	5
7	Worker-Based Numerical Flux Model	5
7.1	Disk discretisation	5
7.2	Why numerical quadrature is used	5
8	Finite Exposure-Time Integration	6
9	Circular and Eccentric Geometry	6
9.1	Circular geometry	6
9.2	Approximate eccentric geometry	6
10	Starspot Morphology Mode	7
10.1	Starspot controls	7
10.2	Scientific caution	7
11	Exomoon Hypothesis Mode	8
11.1	Exomoon controls	8
12	Scientific Readout Metrics	8
12.1	Brightness dip	8
12.2	Radius proxy	8
12.3	Residual RMS	9
12.4	Out-of-transit RMS	9
12.5	Depth contrast	9
12.6	Model-assumption strip	9

13 Model Controls	9
13.1 Rendering controls	9
13.2 Planet and orbit controls	10
13.3 Stellar atmosphere controls	10
13.4 Model alignment	10
14 WebGL Scene and Visual Synchronisation	10
14.1 Scene versus physics	10
15 Software Architecture	11
15.1 index.html	11
15.2 styles.css	11
15.3 src/app.js	11
15.4 src/transitWorker.js	11
15.5 src/scene.js	12
16 Latest-State Mailbox Architecture	12
16.1 Concept	12
17 Data Format for Users	12
17.1 Target catalogue example	12
17.2 Light-curve object format	13
17.3 Light-curve point-list format	13
17.4 Recommended preprocessing	13
18 Comparison with Professional Platforms	14
19 Performance Notes	14
20 Known Limitations	15
21 Future Roadmap	15
22 Quick Glossary	15
23 Summary for Visitors	16
References	16

1 Scope and Purpose

ExoIntel-Prime is a browser-based exoplanet transit photometry laboratory. It combines a local target archive, cached light curves, an interactive theoretical transit model, a scientific readout panel, and a cinematic WebGL star–planet visualisation. The main goal is to help visitors understand how transit light curves respond to planetary radius, orbital geometry, limb darkening, stellar surface features, finite exposure time, and possible additional occulting bodies.

The current release is designed as an educational and scientific-communication tool, not as a final professional inference pipeline. The interface intentionally separates:

- **archival photometry**, shown as static observed data points;
- **theoretical model flux**, computed by a background solver engine;
- **WebGL visualisation**, rendered in `src/scene.js`;
- **interactive controls**, managed by `src/app.js`;
- **physical assumptions and limitations**, displayed through the readout and model-assumption strip.

Scientific transparency

A close visual match between the model curve and the data does not by itself prove a planet parameter, starspot, exomoon, or binary-system interpretation. ExoIntel-Prime is a transparent browser model for exploration. Formal claims require data provenance, systematics treatment, uncertainty modelling, parameter inference, and independent validation against established tools.

2 What ExoIntel-Prime Does

The website allows a user to:

1. select a target from `data/exoplanets.json`;
2. load a local phase-folded light curve where available;
3. compare archival photometry with a theoretical model;
4. adjust physical and hypothesis parameters using sliders and switches;
5. inspect live scientific metrics such as brightness dip, residual RMS, out-of-transit RMS, depth contrast, starspot anomaly, and moon-signal estimate;
6. visualise the star–planet system in a WebGL viewport;
7. switch between light and night modes;
8. use a cinematic boot screen that introduces the scientific engine before the interface opens.

2.1 Who the site is for

The tool is designed for:

- science-communication visitors who want an intuitive transit demonstration;
- students learning exoplanet photometry;
- GitHub users who want to inspect or fork the code;

- early-stage researchers who want a visual bridge between catalogue parameters and light-curve shape;
- portfolio viewers assessing the technical architecture of the project.

2.2 What the site is not

The tool should not be treated as a replacement for professional packages such as **batman**, **PyTransit**, **allesfitter**, **juliet**, **EXOFAST**, or **starry**. It does not yet perform full posterior inference, detrending, dilution modelling, Gaussian-process stellar variability treatment, or model comparison using Bayesian evidence.

3 Interface Overview

3.1 Launch screen

The public interface begins with a short cinematic launch sequence. This sequence is visual only; it does not change the physics. It shows a staged progress bar from 0 to 100 percent with messages such as:

- *Initialising observatory systems...*
- *Loading the exoplanet target archive...*
- *Preparing the transit physics engine...*
- *Tighten your seatbelt — we are travelling to another star system...*

This improves the presentation of the website while preserving a static, GitHub Pages-compatible architecture.

3.2 Main layout

The application is divided into four scientific zones:

Zone	Purpose
Target archive	Search and select an exoplanet system from the local JSON catalogue.
Scientific readout	Display human-readable depth, ppm, residuals, planet parameters, stellar parameters, and data provenance.
CGI model viewport	Render the current theoretical star–planet configuration using WebGL.
Model controls	Adjust physical/hypothesis parameters for interactive “what-if” exploration.

4 Target Archive and Data Philosophy

Targets are stored locally in `data/exoplanets.json`. Light curves are stored separately in `data/lightcurves/`. The website is intentionally static: visitors cannot modify the production data directly. Users who want to test their own data should fork the repository, add a light-curve JSON file, update `data/exoplanets.json`, and deploy their own copy or submit a pull request.

4.1 Target archive fields

The catalogue may contain the following physical and metadata fields:

Field	Unit	Meaning
pl_name	–	Planet name.
hostname	–	Host star name.
pl_orbper	days	Orbital period.
pl_trandur	hours	Transit duration.
pl_trandep	ppm	Transit depth from catalogue or processed source.
pl_ratror	–	Planet-to-star radius ratio, $R_p/R_\star$.
pl_orbsmax	AU	Semi-major axis.
pl_orbincl	degrees	Orbital inclination.
pl_orbeccen	–	Orbital eccentricity.
pl_orblper	degrees	Argument of periastron, ω , where available.
pl_rade	$R_\oplus$	Planet radius in Earth radii.
pl_radj	R_J	Planet radius in Jupiter radii.
pl_bmasse	$M_\oplus$	Planet mass in Earth masses.
pl_bmassj	M_J	Planet mass in Jupiter masses.
st_teff	K	Stellar effective temperature.
st_rad	$R_\odot$	Stellar radius.
st_mass	$M_\odot$	Stellar mass.
st_logg	cgs	Stellar surface gravity.
st_met	dex	Stellar metallicity, usually [Fe/H].
lightcurve_file	–	Local JSON file in <code>data/lightcurves/</code> .
lightcurve_available	boolean	Whether the target has local photometry.

4.2 Observed versus synthetic targets

A target labelled **observed** has a local light-curve JSON file. A target labelled **model** does not have a local light curve and may use a synthetic fallback curve for demonstration. Synthetic fallback curves are useful for teaching, but they should never be described as archival data.

5 Core Transit Physics

5.1 Transit depth

When a planet passes in front of a star, it blocks part of the stellar disk and causes a decrease in the observed flux. For a small planet crossing a uniform stellar disk, the approximate fractional depth is

$$\delta \simeq \left(\frac{R_p}{R_\star} \right)^2. \quad (1)$$

The radius ratio $k = R_p/R_\star$ is therefore one of the most important controls in the interface. The model readout reports depth both as a percentage and in parts per million:

$$\delta_{\text{ppm}} = 10^6 \delta, \quad \delta_{\%} = 100 \delta. \quad (2)$$

The conversion is

$$10,000 \text{ ppm} = 1\%. \quad (3)$$

Why the website shows percent and ppm

Professional transit photometry often uses ppm because signals are small. General visitors usually understand percentage more easily. ExoIntel-Prime therefore shows both:

$$28,000 \text{ ppm} = 2.8\%.$$

5.2 Orbital phase

The horizontal axis of the light-curve plot is orbital phase,

$$\phi = \frac{t - T_0}{P}, \quad (4)$$

where t is time, T_0 is a reference mid-transit time, and P is the orbital period. In the model convention, phase zero is centred near mid-transit before any user-applied phase shift.

5.3 Scaled orbital distance

The scaled distance

$$a/R_\star = \frac{a}{R_\star} \quad (5)$$

controls the projected orbital speed and transit duration. In the application, if a catalogue semi-major axis and stellar radius are available, the app estimates

$$\frac{a}{R_\star} \simeq \frac{a_{\text{AU}}}{R_{\star,\odot} \times 0.00465047}, \quad (6)$$

because $1R_\odot \simeq 0.00465047 \text{ AU}$.

5.4 Impact parameter

For a circular orbit, the approximate impact parameter is

$$b = \frac{a}{R_\star} \cos i, \quad (7)$$

where i is the orbital inclination. A central transit has $b \approx 0$, while a grazing transit has b close to $1 + k$.

5.5 Transit duration

For a circular orbit, a simplified first-to-fourth contact duration is

$$T_{14} \simeq \frac{P}{\pi} \arcsin \left[\frac{1}{a/R_\star} \frac{\sqrt{(1+k)^2 - b^2}}{\sin i} \right], \quad k = \frac{R_p}{R_\star}. \quad (8)$$

The web app does not rely only on this analytic approximation. The background solver projects the occulting body onto a limb-darkened stellar disk and calculates the blocked intensity numerically.

6 Limb Darkening

Real stellar disks are not uniformly bright. The centre is usually brighter than the limb. ExoIntel-Prime uses the quadratic limb-darkening law:

$$\frac{I(\mu)}{I(1)} = 1 - u_1(1 - \mu) - u_2(1 - \mu)^2, \quad (9)$$

where

$$\mu = \cos \theta = \sqrt{1 - r^2}. \quad (10)$$

Here r is the projected radial coordinate on the stellar disk. The coefficients u_1 and u_2 are controlled by the sliders labelled `Quadratic limb u1` and `Quadratic limb u2`.

6.1 Why limb darkening matters

Limb darkening changes the detailed shape of ingress, egress, and the transit bottom. It can affect inferred values of R_p/R_* , impact parameter, and stellar density. Analytic transit models such as Mandel and Agol [1] include quadratic and nonlinear limb-darkened cases. Kipping [4] introduced a useful reparameterisation for two-coefficient limb-darkening laws to avoid unphysical coefficient combinations during statistical fitting.

Current interface choice

The current public interface exposes u_1 and u_2 directly because they are easier for users to understand. A future fitting-oriented release should offer Kipping's q_1, q_2 parameterisation for physically bounded sampling.

7 Worker-Based Numerical Flux Model

7.1 Disk discretisation

The theoretical flux is calculated in `src/transitWorker.js`. This module functions as the background computation engine and discretises the stellar disk into polar samples. Each sample has a projected position (x_j, y_j) and an intensity weight I_j . The unocculted stellar flux is

$$F_0 = \sum_{j=1}^N I_j. \quad (11)$$

At a given phase, the planet covers a subset of disk samples $\mathcal{O}$. The normalised flux is

$$F(\phi) = 1 - \frac{\sum_{j \in \mathcal{O}} I_j}{\sum_{j=1}^N I_j}. \quad (12)$$

This makes the plotted model directly connected to projected occultation geometry.

7.2 Why numerical quadrature is used

Analytic transit models are faster and more precise for standard cases. Mandel and Agol [1] are the classic reference for analytic light curves, and packages such as `batman` [6] and `PyTransit` [7] are widely used for professional modelling. However, numerical quadrature is flexible and simple to explain in a browser. It also allows the same conceptual engine to include irregular starspot masks and a second occulting body.

Validation status

The current browser model is physically interpretable, but it should be validated against established tools before being advertised as a precision fitting engine. A future validation suite should compare ExoIntel-Prime against **batman**, PyTransit, and analytic Mandel–Agol outputs for controlled test cases.

8 Finite Exposure-Time Integration

Real telescopes do not measure instantaneous flux. They integrate over a finite exposure time. Long integration times can smear ingress and egress and bias retrieved transit parameters if not accounted for [2, 3].

ExoIntel-Prime now includes a finite-exposure integration approximation. Instead of evaluating only

$$F(\phi_i), \quad (13)$$

the background computation engine evaluates several sub-samples across an effective phase width $\Delta\phi_{\text{exp}}$:

$$\bar{F}(\phi_i) = \frac{1}{N_{\text{exp}}} \sum_{m=1}^{N_{\text{exp}}} F(\phi_i + \Delta\phi_m). \quad (14)$$

The effective exposure width is inferred from the median phase spacing of the loaded light curve where possible. If no observed data are present, the background solver uses a conservative fallback based on the orbital period.

What this improves

Finite exposure integration makes the model more realistic for binned or long-cadence photometry. It reduces the risk of drawing unrealistically sharp ingress and egress features.

9 Circular and Eccentric Geometry

9.1 Circular geometry

For circular projected geometry, the background solver uses phase zero near inferior conjunction. In simplified stellar-radius units,

$$x = -a/R_{\star} \sin(2\pi\phi), \quad (15)$$

$$y = (a/R_{\star}) \cos i \cos(2\pi\phi), \quad (16)$$

$$z = (a/R_{\star}) \sin i \cos(2\pi\phi). \quad (17)$$

A positive z value places the planet in front of the stellar disk.

9.2 Approximate eccentric geometry

The upgraded background solver and WebGL scene support approximate eccentric projected geometry using catalogue eccentricity e and argument of periastron ω . The orbital radius is

$$r = \frac{a(1 - e^2)}{1 + e \cos f}, \quad (18)$$

where f is the true anomaly. The background solver resolves Kepler’s equation,

$$M = E - e \sin E, \quad (19)$$

to convert mean anomaly to eccentric anomaly and then to true anomaly.

Important limitation

The eccentric geometry is an interactive projected-orbit approximation anchored near inferior conjunction. It is useful for visual and educational exploration, but it is not yet a full orbital-inference module. A professional eccentric fit should include priors, uncertainties, $e \cos \omega$, $e \sin \omega$, parameter correlations, and validation against standard fitting packages.

10 Starspot Morphology Mode

Starspots are cooler, darker regions on the stellar surface. When a planet crosses a dark spot, it blocks less light than expected from the surrounding photosphere, creating a temporary upward anomaly relative to a spotless transit model.

In ExoIntel-Prime, the spot-modified intensity is represented as

$$I_j^{\text{spot}} = I_j S(x_j, y_j), \quad (20)$$

where

$$S(x, y) = 1 - C M(x, y). \quad (21)$$

Here C is the spot contrast and $M(x, y)$ is an irregular deterministic mask. The spot is intentionally not a perfect circle because a circular blob looks visually artificial and does not communicate stellar-surface complexity well.

10.1 Starspot controls

Control	Meaning
Enable starspot	Turns the spot hypothesis term on or off.
Spot x-position	Moves the spot horizontally across the projected stellar disk.
Spot y-position	Moves the spot vertically across the projected stellar disk.
Spot radius	Changes the approximate spot size.
Spot contrast	Changes how dark the spot is relative to the local photosphere.

10.2 Scientific caution

The starspot mode is not a full stellar activity model. Detailed spot inference can require stellar rotation, repeated transits, spot evolution, multi-band photometry, and surface-map modelling. Tools such as `starry` can represent surface maps analytically using spherical harmonics [8]. ExoIntel-Prime’s current starspot mode is best understood as an interactive morphology demonstration.

11 Exomoon Hypothesis Mode

The exomoon mode adds a second small occulting body near the planet. The approximate moon-only transit depth is

$$\delta_m \simeq \left(\frac{R_m}{R_\star} \right)^2. \quad (22)$$

The metric labelled **Moon signal** reports the maximum moon-only contribution in ppm, if the moon hypothesis is enabled.

11.1 Exomoon controls

Control	Meaning
Enable exomoon	Turns the second occulting body on or off.
Moon radius	Changes the moon radius relative to the host star.
Moon distance	Changes the projected planet–moon separation scale.
Moon phase	Rotates the moon around the planet in the model geometry.

No exomoon detection claim

A moon-like extra dip in a single browser model is not evidence for an exomoon. Real exomoon searches require multi-transit consistency, timing and duration analysis, dynamical constraints, and careful false-positive rejection [9, 10].

12 Scientific Readout Metrics

12.1 Brightness dip

The readout labelled **Brightness dip** is the maximum modelled flux loss:

$$\delta_{\text{model}} = 1 - \min_{\phi} F(\phi). \quad (23)$$

It is shown as both percentage and ppm:

$$\delta_{\%} = 100\delta_{\text{model}}, \quad \delta_{\text{ppm}} = 10^6\delta_{\text{model}}. \quad (24)$$

12.2 Radius proxy

The readout labelled **Radius proxy** uses

$$\frac{R_p}{R_\star} \approx \sqrt{\delta_{\text{model}}}. \quad (25)$$

This approximation is exact only for a uniform stellar disk with no dilution, no blending, and a fully interior transit. With limb darkening, grazing geometry, third-light contamination, and activity, it becomes only a useful intuition.

12.3 Residual RMS

Residual RMS is

$$\text{RMS}_{\text{res}} = 10^6 \sqrt{\frac{1}{N} \sum_{i=1}^N (F_i^{\text{data}} - F_i^{\text{model}})^2} \quad [\text{ppm}]. \quad (26)$$

A smaller value means the current theoretical curve follows the plotted data more closely. It does not prove uniqueness.

12.4 Out-of-transit RMS

Out-of-transit RMS estimates the scatter outside the transit window:

$$\text{RMS}_{\text{OOT}} = 10^6 \sqrt{\frac{1}{N_{\text{OOT}}} \sum_{i=1}^{N_{\text{OOT}}} (F_i - \tilde{F}_{\text{OOT}})^2}. \quad (27)$$

Here $\tilde{F}_{\text{OOT}}$ is the median out-of-transit flux.

12.5 Depth contrast

The readout labelled **Depth contrast** is

$$\text{Depth contrast} \simeq \frac{\delta_{\text{model,ppm}}}{\text{RMS}_{\text{OOT,ppm}}}. \quad (28)$$

This is a quick visual diagnostic. It should not be presented as a formal detection signal-to-noise ratio unless the data processing, noise model, and transit window are carefully defined.

12.6 Model-assumption strip

The small assumption strip under the plot reports items such as:

- preview or high-accuracy numerical quadrature;
- disk quadrature resolution;
- quadratic limb darkening;
- circular or eccentric geometry;
- exposure integration;
- archival photometry loaded or synthetic fallback;
- moon/starspot hypothesis flags.

This is intentionally shown in the interface to keep the model transparent.

13 Model Controls

13.1 Rendering controls

Control	Meaning
---------	---------

Visual quality	Changes the WebGL scene only. It does not change the background solver's physical light-curve calculation. Use Ultra for screenshots and Balanced for normal use.
----------------	---

13.2 Planet and orbit controls

Control	Scientific role
Radius ratio R_p/R_*	Dominant control for transit depth. Larger planets produce deeper transits.
Scaled distance a/R_*	Changes the projected orbital scale and duration behaviour.
Inclination	Controls whether the transit is central, grazing, or absent.
Catalogue eccentricity	Read-only catalogue context passed to the background solver when available.

13.3 Stellar atmosphere controls

Control	Scientific role
Quadratic limb u1	First coefficient of the quadratic limb-darkening law.
Quadratic limb u2	Second coefficient controlling curvature of the intensity profile.

13.4 Model alignment

The phase-shift slider moves the theoretical model relative to the archival phase axis:

$$\phi_{\text{model}} = \phi_{\text{observed}} - \Delta\phi. \quad (29)$$

This is useful when the light curve is slightly offset from the assumed catalogue mid-transit time. It should not be overused to hide a genuinely poor model.

14 WebGL Scene and Visual Synchronisation

The WebGL scene in `src/scene.js` renders a cinematic star-planet system. The star uses procedural shader elements to suggest limb darkening, granulation, coronal glow, and irregular starspot appearance. The planet and moon are shaded bodies whose projected motion follows the same broad geometry convention as the background solver.

14.1 Scene versus physics

The WebGL scene is a visual representation, not the source of the flux calculation. The source of the plotted theoretical flux is the background solver. However, the scene and the background solver now share:

- the same circular/eccentric geometry convention;
- the same catalogue eccentricity and argument of periastron fields;
- the same planet, moon, and starspot control state;
- the same selected target metadata.

Why this matters

Earlier versions of the project risked looking visually attractive while being physically disconnected from the plotted curve. The upgraded architecture reduces that disconnect: the plot and the 3D scene now respond to the same physical state vector.

15 Software Architecture

The current static repository is organised as:

```
exolight-transit-lab/  
|-- index.html  
|-- styles.css  
|-- src/  
|   |-- app.js  
|   |-- scene.js  
|   |-- transitWorker.js  
|-- data/  
|   |-- exoplanets.json  
|   |-- lightcurves/  
|-- docs/
```

15.1 index.html

The HTML entry point defines the boot screen, metadata, page title, and script entry. It is intentionally small so the application remains easy to deploy on GitHub Pages.

15.2 styles.css

The global stylesheet controls the boot screen, default page background, no-script fallback, scrollbars, and accessibility focus states. The main application styles are injected by `src/app.js`.

15.3 src/app.js

The main thread orchestrator handles:

- cinematic boot sequence;
- target loading;
- UI rendering;
- theme switching;
- plot drawing;
- asynchronous solver communication;
- latest-revision management;
- discarding stale solver results.

15.4 src/transitWorker.js

The background computation engine handles:

- numerical transit model calculation;
- finite exposure integration;

- circular/eccentric projected geometry;
- starspot-modified intensity maps;
- exomoon hypothesis contribution;
- residual and out-of-transit diagnostics;
- latest-state mailbox execution.

15.5 src/scene.js

The WebGL renderer handles:

- star shader;
- coronal glow shell;
- planet and moon rendering;
- orbit-path rendering;
- starfield;
- visual quality modes;
- synchronisation with the current model state.

16 Latest-State Mailbox Architecture

Interactive sliders can produce many parameter updates per second. If each update were solved in order, the background computation engine would build a queue and the interface would feel delayed. ExoIntel-Prime instead uses a latest-state mailbox architecture.

16.1 Concept

The main thread attaches a revision number to each parameter snapshot:

$$r = 1, 2, 3, \dots \quad (30)$$

The background computation engine keeps:

- one active solve;
- one pending latest request;
- no long FIFO queue.

If a newer revision arrives while an older solve is still running, the old result is discarded.

Why this improves the experience

The user sees the latest slider state, not a delayed sequence of outdated intermediate models. This is essential for a responsive browser-based scientific interface.

17 Data Format for Users

17.1 Target catalogue example

```
{
  "pl_name": "HD 189733 b",
  "hostname": "HD 189733",
  "discoverymethod": "Transit",
  "disc_year": 2005,

  "pl_orbper": 2.21857567,
  "pl_ratror": 0.15534,
  "pl_trandep": 24000,
  "pl_trandur": 1.823,
  "pl_orbincl": 85.71,
  "pl_orbeccen": 0.0,
  "pl_orblper": 90.0,

  "pl_rade": 12.7,
  "pl_bmasse": 363.0,

  "st_teff": 5052,
  "st_rad": 0.75,
  "st_mass": 0.79,
  "st_logg": 4.55,
  "st_met": -0.03,

  "lightcurve_file": "hd-189733-b.json",
  "lightcurve_available": true
}
```

17.2 Light-curve object format

```
{
  "source": "local processed light curve",
  "phase": [-0.050, -0.049, -0.048],
  "flux": [1.0002, 0.9999, 1.0001],
  "error": [0.0004, 0.0004, 0.0004]
}
```

17.3 Light-curve point-list format

```
{
  "source": "local processed light curve",
  "points": [
    {"phase": -0.050, "flux": 1.0002, "error": 0.0004},
    {"phase": -0.049, "flux": 0.9999, "error": 0.0004}
  ]
}
```

17.4 Recommended preprocessing

Before adding a custom light curve:

1. remove NaN, infinite, and physically impossible values;
2. normalise the out-of-transit baseline close to 1;
3. convert time to orbital phase using the expected period and mid-transit time;

4. remove or flag severe outliers;
5. record the source and processing notes;
6. keep the JSON file small enough for static hosting;
7. avoid overwriting the production archive unless submitting a reviewed pull request.

18 Comparison with Professional Platforms

ExoIntel-Prime is not intended to replace professional data archives. Instead, it fills a different niche.

Platform type		Strength	How ExoIntel-Prime differs
NASA Archive	Exoplanet	Authoritative catalogue data and confirmed/candidate planet tables.	ExoIntel-Prime focuses on interactive visual explanation and local static deployment.
MAST		Mission data access, including Kepler, K2, TESS, and HST products.	ExoIntel-Prime uses cached/local light curves rather than acting as a full data portal.
DACE-style platforms	analysis	Professional exoplanet data exploration and analysis workflows.	ExoIntel-Prime emphasises browser-based education, visual intuition, and public accessibility.
batman/PyTransit		Fast and accurate transit model generation.	ExoIntel-Prime uses a browser numerical model for transparency and interaction.
starry		Analytic occultation and surface mapping.	ExoIntel-Prime has a simplified irregular starspot demonstration, not full map inference.

Best positioning

The strongest public positioning is not “we replace MAST or DACE.” It is: *ExoIntel-Prime bridges archival transit data, physical intuition, and interactive visual modelling in a fully static browser application.*

19 Performance Notes

High-quality WebGL rendering and dense numerical modelling can use CPU/GPU resources. Recommended use:

- use **Balanced** visual quality for normal work;
- use **Ultra** for screenshots and short demonstrations;
- use **High-accuracy model** after choosing a target and approximate parameters;
- use **Low** visual quality on older laptops or battery power;
- if the browser becomes warm, reduce visual quality first because the physics engine is already isolated from rendering.

The visual-quality selector affects only the WebGL scene. It does not change the physical model unless the high-accuracy physics button is pressed.

20 Known Limitations

The current public release has several intentional limitations:

1. **No formal fitting:** the app does not yet optimise parameters using least squares, MCMC, or nested sampling.
2. **No posterior uncertainties:** reported values are deterministic readouts, not parameter constraints.
3. **Simplified starspots:** the spot morphology is educational and visual, not a full stellar surface reconstruction.
4. **Exomoon mode is illustrative:** it does not search for or validate exomoons.
5. **No dilution/blending model:** third light from nearby stars is not yet included.
6. **Limited detrending:** local light curves are assumed to be preprocessed.
7. **Approximate eccentric geometry:** useful for visual exploration, but not a substitute for full eccentric transit fitting.
8. **No mission-specific systematics model:** TESS, Kepler, K2, and ground-based data can contain systematics not handled by the browser model.

21 Future Roadmap

A scientifically stronger future release should include:

- validation against Mandel–Agol, **batman**, and PyTransit test cases;
- optional analytic transit model mode;
- Kipping q_1, q_2 limb-darkening parameterisation;
- uncertainty-weighted residuals using reported flux errors;
- automatic phase alignment;
- dilution/third-light parameter;
- finite exposure controls visible in the UI;
- CSV upload and client-side parsing;
- detrending tools for polynomial, spline, or Gaussian-process baselines;
- formal optimiser mode for best-fit parameters;
- multi-transit stacking;
- model comparison between planet-only, starspot, moon-like, and binary-like hypotheses;
- documentation examples using real TESS/Kepler/K2 light curves.

22 Quick Glossary

Term	Meaning
Transit	A planet passing in front of a star and causing a brightness dip.

ppm	Parts per million. 10,000 ppm = 1%.
Normalised flux	Flux scaled so the out-of-transit baseline is near 1.
$R_p/R_\star$	Planet radius divided by stellar radius. Controls transit depth strongly.
$a/R_\star$	Semi-major axis divided by stellar radius. Affects duration and orbital geometry.
Inclination	Tilt of the orbit relative to the observer. Near 90° favours transits.
Impact parameter	Projected transit chord distance from the stellar centre.
Limb darkening	Stellar disk brightness variation from centre to limb.
Finite exposure integration	Averaging model flux over a finite exposure/bin duration.
Residual RMS	Typical model–data mismatch.
OOT RMS	Out-of-transit scatter.
Depth contrast	Model depth divided by out-of-transit scatter.
Starspot anomaly	Flux anomaly caused when a planet crosses a darker stellar region.
Exomoon hypothesis	Second-occultor demonstration mode; not a detection claim.
Background solver	Asynchronous browser computation module used for physics evaluation away from the UI thread.
WebGL	Browser graphics technology used for the 3D scene.

23 Summary for Visitors

ExoIntel-Prime lets users explore how an exoplanet transit light curve changes when the planet size, orbital geometry, stellar atmosphere, starspots, and possible moon-like occultors are adjusted. The plotted theoretical curve is calculated by a background solver and compared against local archival or demonstration photometry. The 3D scene provides physical intuition, while the scientific readout explains the numerical meaning of the model.

The project is strongest when presented as a transparent, interactive bridge between real exoplanet data and transit physics. It is not yet a precision fitting pipeline, but it is designed in a way that can grow toward more rigorous modelling and validation.

References

References

- [1] Mandel, K. and Agol, E. (2002). *Analytic Light Curves for Planetary Transit Searches*. The Astrophysical Journal Letters, 580, L171–L175.
- [2] Kipping, D. M. (2010). *Binning is sinning: morphological light-curve distortions due to finite integration time*. Monthly Notices of the Royal Astronomical Society, 408, 1758–1769.
- [3] Price, E. M. and Rogers, L. A. (2014). *Transit Light Curves with Finite Integration Time: Fisher Information Analysis*. The Astrophysical Journal, 794, 92.
- [4] Kipping, D. M. (2013). *Efficient, uninformative sampling of limb darkening coefficients for two-parameter laws*. Monthly Notices of the Royal Astronomical Society, 435, 2152–2160.
- [5] Espinoza, N. and Jordán, A. (2015). *Limb darkening and exoplanets: testing stellar model atmospheres and identifying biases in transit parameters*. Monthly Notices of the Royal Astronomical Society, 450, 1879–1899.

- [6] Kreidberg, L. (2015). *batman: BAsic Transit Model cAlculationN in Python*. Publications of the Astronomical Society of the Pacific, 127, 1161–1165.
- [7] Parviainen, H. (2015). *PyTransit: fast and easy exoplanet transit modelling in Python*. Monthly Notices of the Royal Astronomical Society, 450, 3233–3238.
- [8] Luger, R., Agol, E., Foreman-Mackey, D., Fleming, D. P., Lustig-Yaeger, J. and Deitrick, R. (2019). *starry: Analytic Occultation Light Curves*. The Astronomical Journal, 157, 64.
- [9] Kipping, D. M. (2009). *Transit timing effects due to an exomoon*. Monthly Notices of the Royal Astronomical Society, 392, 181–189.
- [10] Kipping, D. M. and Teachey, A. (2020). *Impossible moons: Transit timing effects that cannot be due to an exomoon*. Monthly Notices of the Royal Astronomical Society, 497, 2885–2901.
- [11] Giménez, A. (2006). *Equations for the analysis of the light curves of extra-solar planetary transits*. Astronomy and Astrophysics, 450, 1231–1237.
- [12] Eastman, J., Gaudi, B. S. and Agol, E. (2013). *EXOFAST: A Fast Exoplanetary Fitting Suite in IDL*. Publications of the Astronomical Society of the Pacific, 125, 83–112.
- [13] Aigrain, S., Parviainen, H. and Pope, B. J. S. (2016). *K2SC: flexible systematics correction and detrending of K2 light curves using Gaussian process regression*. Monthly Notices of the Royal Astronomical Society, 459, 2408–2419.
- [14] Feinstein, A. D. et al. (2019). *eleanor: An open-source tool for extracting light curves from the TESS Full Frame Images*. Publications of the Astronomical Society of the Pacific, 131, 094502.
- [15] Huang, C. X. et al. (2020). *Photometry of 10 Million Stars from the First Two Years of TESS Full Frame Images: Part I*. Research Notes of the AAS, 4, 204.